

100%

Excellent!

PHY201 Exam on chapter(s): 13, 14, 15.

Name: _____

These questions assess your understanding of the material from Chapters 13, 14, 15 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. Credit is only awarded if you answer correctly and clearly show all major steps necessary to find your answer; credit is not awarded for missing work, unclear work, or the use of memorized equations absent from the equation sheet. Half credit may be awarded for insufficient work that suggests some degree of understanding. Half credit is awarded for correct numerical answers with incorrect or missing units.

1. Suppose that there is heat transfer of 45 J to a system, while the system does 45 J of work. Later, there is heat transfer of 30 J out of the system while 46 J of work is done on the system. What is the net change in internal energy of the system?

ANSWER: 16 J ✓

$$\Delta U = Q - W = \Sigma Q - \Sigma W$$

$$\Delta U = (45 \text{ J} - 30 \text{ J}) - (45 \text{ J} - 46 \text{ J}) = \boxed{16 \text{ J}}$$

2. Benjamin puts a 0.37-kg aluminum pot on the stove and fills it with 0.12 liters of water. Both the pot and the water are initially at 7.1°C. How much heat is required to bring them both to 67.8°C? The specific heat value of water is 4186 J/kg°C and for aluminum it is 900 J/kg°C.

ANSWER: 50703.92 J ✓

$$Q_T = Q_A + Q_W = m_A c_A \Delta T + m_W c_W \Delta T$$

$$\rho_W = \frac{m_W}{V_W} \rightarrow m_W = V_W \rho_W$$

$$Q_T = \Delta T (m_A c_A + V_W \rho_W c_W)$$

$$Q_T = (67.8^\circ\text{C} - 7.1^\circ\text{C}) \left((0.37 \text{ kg})(900 \frac{\text{J}}{\text{kg}\cdot^\circ\text{C}}) + (0.00012 \text{ m}^3)(1000 \frac{\text{kg}}{\text{m}^3})(4186 \frac{\text{J}}{\text{kg}\cdot^\circ\text{C}}) \right) = \boxed{50703.924 \text{ J}}$$

3. Chloe inflates her bicycle tire to an absolute pressure of 0.93 MPa (a gauge pressure of about 120 psi) at a temperature of 7.3°C. What is the tire's gauge pressure after the temperature has risen to 36.7°C? Assume there are no leaks and the volume of the tire does not change.

ANSWER: 926.17 kPa ✓

$$\frac{P_1}{T_1} = \frac{P_2}{T_2} = \frac{P_g + P_{atm}}{T_2} \rightarrow P_g = \frac{T_2 P_1}{T_1} - P_{atm}$$

$$P_g = \frac{(36.7 + 273.15) \text{ K} (930 \text{ kPa})}{(7.3 + 273.15) \text{ K}} - 101.325 \text{ kPa} = \boxed{926.17 \text{ kPa}}$$

+3

4. Nicholas drives a 11300-kg truck and descends a 23-m-tall hill at a constant speed. What is the final temperature of 220 kg of brake material if its temperature is 8°C at the top of the hill? Assume the specific heat of the brake material is 850 J/kg°C. Assume the brakes do all the work to maintain the speed.

$$Q = m_b c \Delta T = m_b c (T_f - T_i) = P E_g = m_T g h$$

$$T_f = \frac{m_T g h}{m_b c} + T_i$$

$$T_f = \frac{(11300 \text{ kg})(9.8 \frac{\text{m}}{\text{s}^2})(23 \text{ m})}{(220 \text{ kg})(850 \frac{\text{J}}{\text{kg} \cdot ^\circ\text{C}})} + 8^\circ\text{C} = \boxed{21.62^\circ\text{C}}$$

ANSWER: 21.62 °C

5. Calculate the work output of a Carnot engine operating between temperatures of 687 K and 333 K for 3550 J of heat transfer to the engine.

$$\eta_c = 1 - \frac{T_c}{T_h} = \frac{W}{Q_h} \rightarrow W = Q_h \left(1 - \frac{T_c}{T_h} \right)$$

$$W = (3550 \text{ J}) \left(1 - \frac{333 \text{ K}}{687 \text{ K}} \right) = \boxed{1829.26 \text{ J}}$$

ANSWER: 1829.26 J

6. Sarah is producing heat at a rate of about 118 W while doing light exercise. At what rate must water evaporate from her body to get rid of the produced energy? The latent heat of vaporization of water is 2430 kJ/kg at body temperature, 37°C.

$$Q = mL_v \rightarrow \frac{Q}{t} = \frac{m L_v}{t} = P \rightarrow \frac{m}{t} = \frac{P}{L_v}$$

$$\frac{m}{t} = \frac{118 \frac{\text{J}}{\text{s}}}{2430000 \frac{\text{J}}{\text{kg}}} = \boxed{4.86 \times 10^{-5} \frac{\text{kg}}{\text{s}}}$$

ANSWER: $4.86 \times 10^{-5} \text{ kg/s}$

7. Spontaneous heat transfer from hot to cold is an irreversible process. Calculate the total change in entropy if 6890 J of heat transfer occurs from a hot reservoir at 534 K to a cold reservoir at 275 K, assuming that there is no temperature change in either reservoir.

ANSWER: $12.15 \frac{\text{J}}{\text{K}}$

$$\Delta S_T = \Delta S_H + \Delta S_C = \frac{Q_h}{T_h} + \frac{Q_c}{T_c}$$

$$\Delta S_T = \frac{-6890 \text{ J}}{534 \text{ K}} + \frac{6890 \text{ J}}{275 \text{ K}} = \boxed{12.15 \frac{\text{J}}{\text{K}}}$$

8. What is the average kinetic energy of a gas molecule at 21°C?

ANSWER: $6.09 \times 10^{-21} \text{ J}$

$$\overline{KE} = \frac{3}{2} k_B T$$

$$\overline{KE} = \frac{3}{2} (1.38 \times 10^{-23} \frac{\text{J}}{\text{K}}) (21 + 273.15) \text{ K} = \boxed{6.09 \times 10^{-21} \text{ J}}$$

9. Find the increase in entropy as 3.8 kg of ice, originally at 0.3°C, melts to form water at 0.3°C at a particular pressure. The latent heat of fusion for water is 334 kJ/kg.

ANSWER: $4.64 \frac{\text{kJ}}{\text{K}}$

$$\Delta S = \frac{Q}{T} = \frac{mL_f}{T}$$

$$\Delta S = \frac{(3.8 \text{ kg})(334 \frac{\text{kJ}}{\text{kg}})}{(0.3 + 273.15) \text{ K}} = \boxed{4.64 \frac{\text{kJ}}{\text{K}}}$$

10. Jennifer's 59-L (15.6-gal) steel gasoline tank is full of gas. Both the tank and the gasoline have an initial temperature of 6.2°C. After both the tank and the gasoline warm to 36.8°C, what is the pressure in the gasoline tank if the gasoline cannot spill out? The coefficient of volume expansion for steel is $35 \times 10^{-6} \text{ } ^\circ\text{C}^{-1}$ and for gasoline it is $950 \times 10^{-6} \text{ } ^\circ\text{C}^{-1}$. Assume the bulk modulus B for gasoline is $1.00 \times 10^9 \text{ N/m}^2$. Show all work.

$$\Delta V = \frac{1}{B} \frac{F}{A} V_i = \Delta V_L = \frac{1}{B} P V_i = \Delta V_G - \Delta V_S = \beta_G V_i \Delta T - \beta_S V_i \Delta T$$

$$P = \frac{\beta_G V_i \Delta T (B_G - B_S)}{V_i} = B \Delta T (\beta_G - \beta_S)$$

$$P = (1 \times 10^9 \frac{\text{N}}{\text{m}^2}) (36.8^\circ\text{C} - 6.2^\circ\text{C}) (950 \times 10^{-6} \frac{1}{^\circ\text{C}} - 35 \times 10^{-6} \frac{1}{^\circ\text{C}}) = 279.99 \times 10^5 \text{ Pa}$$

can represent as

$$28 \times 10^6 \text{ Pa}$$

$$28 \text{ MPa}$$

11. Anna wants to use her pressure cooker to cook food in liquid water at a temperature of 161°C. To determine the pressure required inside the pressure cooker for water to boil at this temperature, she can use the Clausius-Clapeyron equation for water: $P(T) = (1 \text{ kPa}) \cdot \exp \left[(4750 \text{ K}) \left(\frac{1}{273.15 \text{ K}} - \frac{1}{T} \right) \right]$ In this equation, T is the temperature in Kelvin and P is the pressure in kPa. Given that the lid of the pressure cooker is a disk with a diameter of 29 cm, what is the force that the lid must be able to withstand at the required pressure?

$$P_g = \frac{F}{A} = \frac{F}{\pi \left(\frac{d}{2} \right)^2} = P - P_{\text{atm}} = P(T) - P_{\text{atm}}$$

$$F = \pi \left(\frac{d}{2} \right)^2 (P(T) - P_{\text{atm}})$$

$$F = \pi \left(\frac{0.29 \text{ m}}{2} \right)^2 \left((1 \frac{\text{kN}}{\text{m}^2}) e^{(4750 \text{ K}) \left(\frac{1}{273.15 \text{ K}} - \frac{1}{(161+273.15) \text{ K}} \right)} - 101.325 \frac{\text{kN}}{\text{m}^2} \right) = 35.05 \text{ kN}$$

12. Jason's 55-L (14.5-gal) steel gasoline tank is full of gas. Both the tank and the gasoline have an initial temperature of 12.7°C. How much gasoline has leaked out due to thermal expansion by the time both the tank and the gasoline have warmed to 37.1°C? The coefficient of volume expansion for steel is $35 \times 10^{-6} \text{ } ^\circ\text{C}^{-1}$ and for gasoline it is $950 \times 10^{-6} \text{ } ^\circ\text{C}^{-1}$. Show all work.

$$\Delta V_L = \Delta V_G - \Delta V_S = \beta_G V_i \Delta T - \beta_S V_i \Delta T = V_i \Delta T (\beta_G - \beta_S)$$

$$\Delta V_L = (55 \text{ L}) (37.1^\circ\text{C} - 12.7^\circ\text{C}) (950 \times 10^{-6} \frac{1}{^\circ\text{C}} - 35 \times 10^{-6} \frac{1}{^\circ\text{C}}) = 1.23 \text{ L}$$